Ofqual QAN	**600/3055/0**
Quartz ID	**1158 / 1159 / 1161**
Qualification	**Level 5 Diploma in Computing / Level 5 Diploma in Computing with Business Management / Level 5 Diploma in Computing with Data Science**

Assessment	**Introduction to Data Science and Big Data**
Assessment Type	**Examination**
Date	**Tuesday 4th March 2025**
Time	**09.00am – 12.00pm**

Examination Paper

Answer ALL questions.

Clearly cross out surplus answers.

Time: 3 Hours

The maximum mark for this paper is 100.

Any reference material brought into the examination room must be handed to the invigilator before the start of the examination.

Answer ALL questions

Marks

Question 1

a) Define big data **and** explain how it differs from traditional databases. **5**

Discuss the key characteristics of big data.

b) Provide an example of how big data is being used in healthcare or finance. **5**

Total 10 Marks

Question 2

a) Explain the steps involved in the data science lifecycle. **5**

b) Discuss the importance of pre-processing in the data science lifecycle. **5**

Provide examples of how errors in the steps in a) can impact project outcomes.

Total 10 Marks

Question 3

a) What are categorical and continuous data types? Provide examples of each. **5**

b) Explain the concept of Principal Component Analysis (PCA) and its importance in data analysis. **5**

Provide an example of when PCA might be used.

Total 10 Marks

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Marks

Question 4

a) Describe how missing values occur in a dataset **and** why they are problematic. **5**

b) Discuss **TWO (2)** methods for handling missing values in data analysis. You should identify advantages and disadvantages of each. **5**

Total 10 Marks

Question 5

a) Explain in detail the difference between exploratory data analysis **and** confirmatory data analysis. **6**

b) Provide an example with explanation where **both** approaches in a) would be applied in a data science project. **4**

Total 10 Marks

Question 6

a) Explain in detail what is meant by a classification model in supervised learning. **5**

b) Discuss the importance of model evaluation in ensuring model accuracy. **5**

Total 10 Marks

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Marks

Question 7

a) Discuss the challenges of visualising multi-dimensional data **and** explain which types of plots or charts can help to overcome these challenges. **6**

b) Discuss **TWO (2)** examples where advanced visualisation techniques for multi-dimensional data are essential for decision-making **and** explain why these techniques are effective in those scenarios. **4**

Total 10 Marks

Question 8

a) Discuss the importance of evaluation metrics in model scoring. **10**

Identify **and** explain **FOUR (4)** evaluation metrics used to assess model performance **and** provide examples of when each would be most appropriate to use.

Total 10 Marks

Marks

Question 9

a) Explain the role of advanced BI (Business Intelligence) tools in predictive analytics **and discuss** how they contribute to proactive decision-making in industries like finance or healthcare. **10**

Total 10 Marks

Question 10

a) Discuss **ONE (1)** ethical issue related to data aggregation in big data projects. **5**

b) Explain how organisations ensure accountability **and** transparency in the use of big data? **5**

Total 10 Marks

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End of paper

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